

RITHIK BALA

Chennai, India

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EDUCATION

BTech. Mechatronics and Automation (2023-2027) | Vellore Institute of Technology, Chennai

Current Year: 3rd year | **CGPA:** 8.75/10

SKILLS

Mechanical Design & CAD: SolidWorks (CSWP), Fusion 360.

Embedded Systems & Electronics: C/C++, Arduino, ESP8266, Altium, LT Spice, Keil µVision 5.

Programming & Computation: Python, Java, MATLAB.

Robotics & Simulation: Unity (Digital Twins), Arduino IDE, ROS Gazebo.

PATENTS AND PUBLICATIONS

System and Method for Authenticating Pharmaceutical Product

[Aug 2025]

- Designed a system embedding unique QR codes on individual tablet packaging for real-time drug authentication and traceability.
- Integrated QR-linked secure database access to retrieve medicine details, verify authenticity, and provide real-time safety alerts.

CERTIFICATIONS

Certified SolidWorks Professional (CSWP) | Dassault Systèmes

[Dec 2025]

- Advanced CAD Design & Mechanical Modelling

Certified SolidWorks Associate (CSWA) | Dassault Systèmes

[Dec 2025]

- Fundamentals of 3D Modelling & Design Validation

EXPERIENCE

Team Captain | ATOM Robotics (Special Team, VIT Chennai)

[Aug 2025 - Present]

- Leading a team of 30+ members in designing and developing autonomous robotic systems, managing mechanical design, electronics integration, and control implementation.
- Coordinating team efforts and project timelines to deliver functional prototypes for competitions and research.

Discipline Lead | Dance Club (VIT Chennai)

[Aug 2025 - Present]

- Managed discipline and coordination for events with 500+ participants, ensuring structured execution.
- Organized team formations, scheduling, and backstage operations to maintain structured performance flow during events.

Mechanical Engineer | ATOM Robotics (Special Team, VIT Chennai)

[July 2023 - May 2025]

- Contributed to the mechanical design and assembly of Autonomous Race Bots, a 2-DOF robotic arm, and a hexapod robot, resulting in prototypes that achieved over 20% improved mechanical stability in testing.
- Assisted with wiring and firmware calibration, improving hexapod motion accuracy by 15%, leading to smoother and more reliable locomotion.

Technical Department Member | Robotics Club

[Oct 2023 - May 2025]

- Guided over 80 participants in Arduino programming and troubleshooting during Rev Up Arduino workshops, enhancing their practical skills and event engagement.
- Constructed and demonstrated a voice-controlled robotic RC car, successfully showcasing wireless control and system integration to an audience of 100+ attendees.
- Supported assembly and testing of a Soccer Bot prototype, contributing to a 30% increase in operational reliability through hardware and software coordination.

Merchandise Department | Member VIBRANCE'25 (Cultural Fest)

[Feb 2025]

- Managed inventory of 500+ merchandise units during VIBRANCE'25, avoiding stockouts and ensuring timely availability at all distribution points.
- Coordinated planning and execution for Dancepacito'25, facilitating participation of 15 teams and enhancing event flow through efficient logistics.

PROJECTS

Delta Manipulator with VR-Based Digital Twin Controller

[Jan 2026 - Present]

- Developing a desktop-scale delta robot manipulator with precise parallel kinematics for high-speed positioning.
- Building a digital twin in Unity for real-time visualization and control of the physical robot.

Smart Retail Weigh-Fill Packaging System

[Jan 2026 - Present]

- Designing an automated packaging system using load cell sensors for accurate real-time weight measurement.
- Integrating a machine vision system to identify objects, compute total weight, and automatically generate billing.

Conveyor Belt Using Geneva Mechanism

[Aug 2025 - Nov 2025]

- Designed and fabricated a Geneva drive system for intermittent motion and precise indexing.
- Optimized slot geometry and drive mechanism for smooth engagement and reduced mechanical wear.

Hexapod

[Jan 2025 - Mar 2025]

- Constructed a 6-legged hexapod robot with 18 degrees of freedom, featuring a custom-designed 3D-printed chassis optimised for servo placement and structural balance.
- Programmed coordinated wave gait locomotion using Arduino Nano and PCA9685 servo driver, enabling smooth, biomimetic movement with serial-controlled directional input and calibrated servo precision.

Cosmo Clench-RC Controlled Car with Robotic Arm

[Sep 2024 - Oct 2024]

- Fabricated an RC-controlled car with an onboard robotic arm for Cosmo-Clench 24 at IIT Bombay, leveraging a custom metal chassis for mechanical strength.

- Deployed wireless control via ESP8266 and an MDD10A motor driver, allowing precise remote operation of both vehicle and arm for functional task execution.

Non-Contact Tachometer

[Aug 2024 – Oct 2024]

- Designed a non-microcontroller-based tachometer to measure RPM, RPS, and frequency, using pure hardware logic as part of an Analog and Digital Electronics course project.
- Integrated ICs 555, 40110, 4017, 4060 and 4093, along with a TCRT5000 IR sensor, to build a signal processing circuit and display real-time data on a 4-digit 7-segment display.

Voice Controlled Car

[Oct 2023 – Dec 2023]

- Created a voice-controlled robotic car using Arduino UNO, dual-shaft DC motors, and an L293D motor driver, enabling hands-free motion control.
- Developed a custom Android app using MIT App Inventor, implementing Bluetooth-based voice recognition for real-time directional control through spoken commands.

Communicable Disease Prevention Cap (CDPC)

[July 2021]

- Invented a smart wearable cap to promote public health by combining body temperature monitoring with proximity detection for social distancing.
- Integrated a digital temperature sensor with an LCD to display real-time readings in Celsius and Fahrenheit and programmed ultrasonic sensing with buzzer alerts to maintain safe distancing.

ACHIEVEMENTS

- Secured Runner-Up position in the Synthetix 3.0 Hackathon conducted by HumanoidX club, VIT Chennai. **(2025)**
- Participated in IIT Bombay Techfest Zonal, held in Bangalore. **(2024)**
- Secured Third Position in Kalam in You Zonal Conducted By KIOT **(2022)**
- Participated in YESIST'12 in the Junior Einstein Track Conducted By IEEE **(2022)**
- Ranked in the Top 500 Nationwide for the INSPIRE-MANAK award by the Department of Science & Technology, Government of India, and received a ₹10,000 cash prize **(2021)**